

MODULE 1

Automated KYC and Cross-Border Digital Onboarding

Simple learner version with jargon explained

Duration	60 minutes
Level	Apply and analyse
Tools referenced	Tazama, Llama 3, Mistral
Assessment	3 knowledge check questions

How this Word version is structured

This document keeps the same one hour module flow as the original HTML module. Jargon is explained before it is used. The language is simpler, but the compliance meaning is preserved.

The Identity Problem at Scale

Before we start, let us make the difficult words easy. KYC means Know Your Customer. It is the process a bank or financial company uses to confirm who a customer is before doing business with them. Onboarding means accepting a new customer. Cross border onboarding means accepting a customer whose company, owners, documents, or business activity involve more than one country.

Key explanation

Key terms before you begin: KYC: Know Your Customer. The checks used to confirm customer identity and risk. UBO: Ultimate Beneficial Owner. The real human being who finally owns or controls a company. OCR: Optical Character Recognition. Software that reads text from scanned documents or images. LLM: Large Language Model. An AI model that can read text, understand meaning, and produce structured answers. GDPR: General Data Protection Regulation. The law that protects personal data. PII: Personally Identifiable Information. Any data that can identify a person, such as a passport number, address, or date of birth.

Every regulated financial institution has the same problem. It must know exactly who it is dealing with, but modern company documents are long, messy, and often spread across different countries. A company may be owned by another company, which may be controlled by a trust, which may finally benefit one real person. That final real person is the one compliance teams must identify.

Old document systems were built for simple paperwork. They work best when a document has a fixed layout. But international company files rarely look the same. They may include registry documents, shareholder agreements, trust documents, scanned forms, and legal addendums. This creates a serious risk: the first page may look clean, while the real owner is hidden deep inside another document.

Opening Scenario · Aegis Global Logistics Ltd | The Real Owner Hidden Behind Company Layers

A European investment bank receives an onboarding application from Aegis Global Logistics Ltd, registered in Luxembourg. The first company document says that 100% of the shares are held by Vanguard Holding Group LLC, a company registered offshore. The old OCR system reads only the first document and records Vanguard as the shareholder. It gives the file a KYC pass. But another 45 page agreement contains the important detail: Vanguard is connected to a family trust, and Nathaniel Vance holds a 60% beneficial interest in that trust. Nathaniel is politically sensitive and should have triggered extra checks. The old system missed him because it did not understand the full document set.

What This Scenario Means

This is not just a mistake by one person. It is a system design problem. The old system stopped at the first company layer. It did not read the full file. It did not understand the ownership chain. A better AI based system reads all available documents together and follows the meaning across the full file until it reaches the real human owner.

Learning Objectives

By the end of this one hour module, you will be able to:

- Objective 1 Explain why old OCR systems fail when documents come from different countries and different layouts.
- Objective 2 Trace the UBO, meaning the real human owner, through several company layers.
- Objective 3 Explain why customer documents should not be uploaded to public AI tools.
- Objective 4 Describe a local AI process where data stays inside the company system.

- Objective 5 Apply the 25% ownership rule to both direct and indirect ownership.

Key explanation

Regulatory baseline in simple words: FCA means Financial Conduct Authority, the UK financial regulator. SYSC 6.1 means firms must have strong systems to prevent financial crime. FATF means Financial Action Task Force, an international body that sets standards for fighting money laundering. Recommendation 10 says firms must identify and verify the real customer and the real owner. GDPR Article 5 says personal data must be used carefully and only when necessary. GDPR Article 44 controls when personal data can be sent outside protected regions.

CORE LESSON · EIGHT CONCEPTS

From Old Document Reading to Meaning Based AI

1. Why Old OCR Fails

OCR means Optical Character Recognition. It reads text from scanned documents. Traditional OCR often works by looking for information in fixed places on a page. For example, it may expect the shareholder name to appear in the top right corner. If the document layout changes, the system may miss the information.

This is a major problem in cross border onboarding. A German company document, a Cayman Islands certificate, a UAE commercial registration, and a Luxembourg registry extract may all look different. The old system may read one format well and fail on another. Worse, it may fail silently, meaning it misses something without clearly telling the analyst.

Old OCR System

Reads the first registry page. Finds Vanguard Holding Group LLC. Treats it as the shareholder. Misses the trust agreement. Does not identify Nathaniel Vance as the real human owner. Gives the file a false pass.

AI Meaning Based System

Reads the full document set. Connects the registry page, trust details, and ownership wording. Finds that Nathaniel Vance has a 60% beneficial interest. Flags him as the UBO and triggers extra screening.

2. Why Backlogs Become Compliance Risk

A backlog means files waiting to be reviewed. When OCR fails, files move into manual review. One analyst may have to review many long files each day. Each file may include multiple documents and hundreds of pages. Under pressure, important details can be missed.

The risk is simple. If a firm reviews only a small sample of files, it cannot be sure that all hidden owners were found. Complex ownership structures are exactly where criminals may try to hide. That is why firms need better coverage, not just faster processing.

3. Why Public AI Is Not the Safe Answer

A team may be tempted to upload a client document to a public AI tool and ask it to find the owner. This may feel quick, but it creates a serious data protection risk. Client documents often contain PII, meaning Personally Identifiable Information. Examples include names, addresses, passport details, dates of birth, director records, and signatures.

GDPR means General Data Protection Regulation. GDPR Article 5 says personal data must be used only for a clear and necessary purpose. GDPR Article 44 restricts sending personal data outside protected regions unless safeguards are in place. Uploading client documents to an external AI platform may create an unauthorised data transfer.

Key explanation

Simple rule: Do not put sensitive client documents into public AI tools unless the organisation has formally approved that process, the legal safeguards are in place, and the data protection team has confirmed it is allowed.

4. How Large Language Models Read Differently

LLM means Large Language Model. It is an AI model that reads language and understands relationships between words and ideas. Unlike old OCR, it does not need the owner name to appear in a fixed box. It can read a whole paragraph and understand that one person indirectly controls a company through another company or trust.

Semantic processing means reading for meaning. For example, if page 1 says Vanguard owns Aegis and page 32 says Nathaniel has a 60% interest in the trust behind Vanguard, the AI can connect both points and identify Nathaniel as the real owner.

Entity Recognition

The AI identifies names of people, companies, trusts, and countries in the document.

Ownership Tracing

The AI follows the chain from company to company until it reaches the real human owner.

Structured Output

The AI puts the result into a clear format that systems and auditors can read.

Missing Document Checks

The AI can flag unsigned documents, missing fields, and incomplete addendums for human review.

5. The 25% UBO Rule

UBO means Ultimate Beneficial Owner. This is the real human person who owns or controls the company. In many compliance frameworks, a person who owns or controls more than 25% must be identified as a UBO.

The 25% rule applies to direct and indirect ownership. Direct ownership means the person owns shares in their own name. Indirect ownership means the person controls the company through another company, trust, or structure. If Helena owns 45% directly, she is a UBO. If Julian controls a company that owns 55%, Julian is also a UBO because his control is indirect but still above 25%.

Key explanation

Common mistake: Stopping at the first company name is not enough. Compliance teams must continue until they identify the real human person behind the company.

6. Local AI and Tazama

Tazama is an open source compliance software stack. Open source means the software code is openly available and can be reviewed, used, or adapted. In this module, Tazama is used as an example of a private compliance pipeline. Pipeline simply means a step by step process.

The safest design is local processing. Local processing means the documents stay inside the organisation's own technology environment. The AI model, such as Llama 3 or Mistral, runs inside the company system rather than sending client data to a public platform.

Simple Local Workflow

Step 1: The document enters the local system.

Step 2: OCR reads any scanned pages and turns them into text.

Step 3: The local AI model reads the text and extracts the owner details.

Step 4: The result is checked against a required output format.

Step 5: Every action is logged so the firm has an audit trail.

7. System Prompts That Stop Guessing

A system prompt is the instruction given to the AI before it starts. In compliance, the instruction must be strict. The AI should not guess, add opinions, or use outside knowledge. If something is not found in the document, it should say Not Found.

Hallucination means the AI makes up something that sounds possible but is not proven by the document. This is dangerous in compliance. Deterministic output means the output is controlled, consistent, and follows the same required structure every time.

Plain English System Prompt Example You are a private compliance document reading engine. Use only the text provided in this file. Do not search the internet. Do not guess. Find the legal entity name, country of registration, UBOs above 25%, ownership chain, document status, missing signatures, and missing fields. If a field is not available, write "Not Found". Return the result in a structured format only.

8. The Analyst's Role

AI does not replace the compliance analyst. It changes the analyst's work. The AI reads large volumes of documents and highlights risks. The analyst uses judgement on the difficult cases: hidden ownership, politically sensitive people, missing signatures, high risk countries, and incomplete documents.

The model handles volume. The analyst handles judgement. This is the core message of the module.

[PRODUCTION BLUEPRINT](#) · [CAMTASIA SCREEN CAPTURE GUIDE](#)

Building the Training Video

Camtasia is a screen recording and video editing tool. In this module, it is used to show learners how the private KYC pipeline works on screen. Keep the recording simple. Show what the system does, explain the key words, and avoid unnecessary technical detail.

Key explanation

Before recording: Use only fictional data. Do not show real client names, passports, addresses, signatures, or account details. Make sure the local AI system is running inside the company environment and that no public AI website is used.

Phase 1 Show Local Processing

Camtasia Technique: Zoom and callout

What to record: Show the configuration screen where the AI model is connected locally. Explain that local means the system runs inside the organisation.

Voice over: “This setting shows that the AI model is running inside our own environment. The document does not go to a public website. This protects client data and supports GDPR compliance.”

Phase 2 Show the Messy Document

Camtasia Technique: Split screen

What to record: Put the unstructured document on the left and the AI instruction on the right. Unstructured means the document does not follow a neat form.

Voice over: “This document is long and messy. The owner is not shown in one neat box. The AI must read the full meaning across the document.”

Phase 3 Run the Pipeline

Camtasia Technique: Terminal capture

What to record: Run the document processing command and show the structured result appearing on screen.

Voice over: “The system is reading the full file, finding the owners, checking for missing items, and creating a structured output.”

Phase 4 Explain the Output

Camtasia Technique: Field by field zoom

What to record: Zoom into entity name, country, UBO name, ownership percentage, ownership chain, document status, and missing fields.

Voice over: “This output tells the analyst who the real owner is, how ownership is connected, and what still needs human review.”

Phase 5 Show the Human Review Flag

Camtasia Technique: Red alert callout

What to record: Show an addendum that is not fully executed. Addendum means an extra document added to the main agreement. Unexecuted means it has not been properly completed or approved.

Voice over: “The AI has found a missing completion step. This does not mean the AI approves or rejects the customer. It means the analyst must review this exact issue.”

PRODUCTION BLUEPRINT · HEYGEN AI VIDEO SCRIPT

90 Second Introduction Video

HeyGen is an AI video tool that can create an avatar based video. Avatar means a digital presenter. This video should introduce the problem in a human way before learners enter the main lesson.

Production Configuration

Avatar

Use a professional compliance presenter. Keep the tone calm, serious, and clear.

Background

Use a financial data background. Avoid flashy technology visuals.

Subtitles

Use full captions so learners can read every word.

Pacing

Speak slowly enough for non technical learners to follow.

Full Timestamp Script

00:00 to 00:18 Opening

Voice Over

“Every day, compliance teams receive company onboarding files. Some are simple. Others hide the real owner behind several company layers. Our job is to find the real human person behind the paperwork.”

00:18 to 00:35 Old System Problem

Voice Over

“Older systems often look for information in fixed places on a page. When the document layout changes, the system may miss the owner. Files then pile up for manual review.”

00:35 to 00:57 AI Reading Meaning

Voice Over

“A Large Language Model, or LLM, reads meaning across the whole document. It can connect company names, trust details, ownership percentages, and signatures to show who really controls the company.”

00:57 to 01:13 Data Protection

Voice Over

“But speed is not enough. Client data must stay protected. In this module, the AI runs locally, inside the organisation, so sensitive data is not uploaded to a public tool.”

01:13 to 01:30 Learning Promise

Voice Over

“By the end of this session, you will be able to trace real owners, apply the 25% rule, recognise missing documents, and understand how AI can support compliance safely.”

SIMULATION SANDBOX · APPLY YOUR LEARNING

Scenario: Vanguard Apex Solutions

You are a compliance analyst at a regulated investment firm. Regulated means the firm must follow financial rules set by authorities. Read the scenario and identify the real owners, risk flags, and missing document issue.

Corporate Onboarding File

Vanguard Apex Solutions has two ownership blocks. Helena Vance directly owns 45% of the company. Meridian Corporate Capital owns the remaining 55%. A registry lookup shows that Meridian Corporate Capital is fully funded and controlled by Julian Drax. Julian is under regulatory monitoring, meaning authorities are watching him more closely. The main corporate declaration has signatures from Helena and Julian, but the final addendum for international distribution channels has not been completed by the external legal registry.

Your Analysis Tasks

UBO Task

Identify every natural person above 25%. Natural person means a real human being, not a company.

Ownership Chain Task

Explain how each person owns or controls the company, directly or indirectly.

Document Task

Find the incomplete document issue that needs manual review.

Risk Flag Task

Identify who is being monitored and what extra check this should trigger.

Important point

Analyst note: Helena Vance is a UBO because she owns 45% directly. Julian Drax is also a UBO because he controls Meridian Corporate Capital, which owns 55%. The incomplete addendum must be sent for manual review. Julian's monitoring status should trigger enhanced due diligence, meaning extra checks before approval.

KNOWLEDGE CHECK · THREE ASSESSED QUESTIONS

Test Your Analysis

Select the best answer. Each explanation is written in plain language so you can understand the reasoning immediately.

Question 1 of 3 · UBO Identification

Who must be flagged as Ultimate Beneficial Owners, or UBOs, for Vanguard Apex Solutions?

- A. Helena Vance only, because she owns 45% directly.
- B. Julian Drax only, because he controls Meridian Corporate Capital.
- C. Both Helena Vance and Julian Drax. Helena owns 45% directly, and Julian controls 55% indirectly through Meridian Corporate Capital.
- D. Neither person, because Meridian Corporate Capital is a company and companies cannot be checked further.

Correct answer

C. Both Helena Vance and Julian Drax. Helena owns 45% directly, and Julian controls 55% indirectly through Meridian Corporate Capital.

Both must be flagged. Helena owns 45% directly, which is above the 25% rule. Julian controls Meridian Corporate Capital, which owns 55%, so Julian has indirect control above 25%. A UBO must be a real human person, so the check cannot stop at Meridian Corporate Capital.

Question 2 of 3 · Missing Document Check

Which issue must be sent for manual review before onboarding can continue?

- A. Julian Drax has not signed the main corporate declaration.
- B. The final addendum for international distribution channels has not been completed by the external legal registry.
- C. Meridian Corporate Capital owns too much of the company.
- D. The file does not mention a physical office address.

Correct answer

B. The final addendum for international distribution channels has not been completed by the external legal registry.

The correct issue is the incomplete addendum. An addendum is an extra document added to a main agreement. The scenario says the final addendum has not been completed by the external legal registry. That needs human review. The scenario confirms Julian and Helena signed the main declaration, so option A is wrong.

Question 3 of 3 · Safe AI Instruction

Which instruction is safest for a local AI system reviewing this KYC document?

- A. Search the internet and decide whether Julian Drax has negative media coverage.
- B. Extract the company identity, all UBOs above 25%, ownership chains, addendum status, and missing fields. Use only the provided text. If something is missing, write Not Found. Do not guess.
- C. Write a welcome letter to Helena Vance asking for more documents.
- D. Give a probability score for whether the company may be used for financial crime.

Correct answer

B. Extract the company identity, all UBOs above 25%, ownership chains, addendum status, and missing fields. Use only the provided text. If something is missing, write Not Found. Do not guess.

Option B is safest because it gives the AI a clear extraction task, limits it to the provided document, tells it what to do when information is missing, and stops guessing. Compliance output must be evidence based. The AI should not invent risk scores or search outside sources unless that is part of an approved process.

MODULE RECAP · FIVE POINTS TO CARRY FORWARD

What You Learned

This module is about one practical skill: finding the real human owner behind complex company documents while keeping customer data safe.

- Old OCR reads layout. It looks for information in fixed places. If the document format changes, it may miss important details.
- AI reads meaning. A Large Language Model can connect ownership details across the full document set.
- The 25% rule includes indirect control. A person can be a UBO even if they control the customer through another company or trust.
- Public AI creates data risk. Client documents may contain Personally Identifiable Information, so they should not be uploaded without approved safeguards.
- Strict prompts protect the audit trail. The AI must use only the document, avoid guessing, and return Not Found when information is missing.

Important point

Progression to Module 2: Module 2 will apply the same ideas to anti money laundering transaction monitoring. You will move from reading onboarding documents to detecting suspicious behaviour in transaction activity.